

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIC503 - BRIDGE AND TUNNEL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Scour depth at a bridge site is mainly dependent on:
 - a) Type of bearings
 - b) Flow velocity and soil type
 - c) Type of superstructure
 - d) Width of footpath
- 2 Pipe jacking method is commonly used for:
 - a) Metro tunnels
 - b) Long highway tunnels
 - c) Utility and sewer tunnels
 - d) Railway tunnels
- 3 IRC Class AA loading is generally used for:
 - a) Village roads
 - b) Temporary bridges
 - c) National highways and important bridges
 - d) Footbridges only
- 4 Wing walls are mainly provided for:
 - a) Carrying traffic loads
 - b) Retaining approach embankment
 - c) Supporting girders
 - d) Providing parapet height
- 5 Railway bridges differ from road bridges mainly due to:
 - a) Lower impact factors
 - b) Slack load effect
 - c) Higher axle loads and dynamic forces
 - d) Smaller spans
- 6 Hydraulic forces on a bridge superstructure are mainly due to:
 - a) Traffic flow
 - b) Water current and debris impact
 - c) Centrifugal action
 - d) Earthquake
- 7 Bearings in a bridge are provided to:
 - a) Increase stiffness
 - b) Allow controlled movements and rotation
 - c) Reduce scour
 - d) Support only rail bridges

- 8 Cut-and-cover method is suitable for tunnels built in:
 - a) Deep rock
 - b) Soft soil with shallow cover
 - c) Underwater rock
 - d) Hilly regions
- 9 Muck disposal in deep tunnels is generally done by:
 - a) Trucks only
 - b) Conveyor belts and locomotives
 - c) Manually
 - d) Dumping inside tunnel
- 10 Powder factor in blasting represents:
 - a) Energy lost in drilling
 - b) Explosive energy per unit volume of rock
 - c) Explosive cost
 - d) Vibration rate
- 11 Freeboard in a bridge is defined as:
 - a) Distance between soffit and water surface
 - b) Distance between deck top and flood level
 - c) Distance between riverbed and deck
 - d) Distance between bearings
- 12 Tunnel Boring Machines (TBM) are most suitable for:
 - a) Very short tunnels
 - b) Long tunnels requiring fast and continuous excavation
 - c) Soft soil only
 - d) Blasting-based excavation

Q2. Solve the following.

- A) Explain the criteria for selection of bridge site and alignment. 6
- B) What are the components of bridges and draw with sketches? 6

Q3. Solve the following.

- A) Explain in detail various types of bridge loads as per IRC. Include dead load, live load, impact factor, wind load, centrifugal force, seismic load, and earth pressure. 6
- B) Explain the use of influence lines for maximum forces in members. 6

Q4. Solve Any Two of the following.

- A) Explain types of foundations for bridges with sketch. 6
- B) What is the importance of bridge bearings? state their functions, and factors governing bearing selection. 6
- C) Explain orthotropic plate theory. 6

Q5. Solve Any Two of the following.

- A) Discuss the problems commonly encountered in tunnel construction 6
- B) Explain shallow tunneling and deep tunneling with sketch. 6
- C) Explain tunnel support systems used in soft ground and hard rock tunneling. 6

Q6. Solve Any Two of the following.

- A) Explain TBM application with sketch. 6
- B) Discuss the factors influencing tunnel blast performance. 6
- C) What are the selection criteria for equipment for mucking and transportation? 6

*** End ***